

Weekly Site Report

BLACK ROCK WEEKLY REPORT

Weekly PSD Report
EPIROC SA

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Weekly Service Report – Black Rock Mine Operations

Report type:	Weekly	Submitted to:	Sello Taku and Sipho
Week Date:	02 January 2026	End Date	08 January 2026
Reported By:	Philip Moller		
BEV:			
Service Tech:	Ivan Phiri	Service Tech:	Lukas Serumola (resigned)
Service Tech:	Patricia Bulelwa	Service Tech Assistant:	Patrick Yoko (Batteries)
Specialist:	Frikkie van der Walt (Batteries)		
Field Service:			
Service Tech:	Lucas Slaffa	Service Tech:	Mathews Mosemane
CAS Project:			
Team Leader:	Wisani Shilenge	Service Tech:	Phineas Sibanda
Service Tech:	Julius Manamela	Service Tech:	Joseph Mampa
Certiq Project:			
Team Leader:	Segunda Fernando	Service Tech:	
Mobilaris:			
Team Leader:	Rupert Grobler		

Report Summary

General Summary

The purpose of this report is to give an overview of all the Epiroc activities at Blackrock Mine Operations (BRMO).

BEV - Critical Breakdown Summary

- DT171 - A-frame that was previously damaged failed again. A-frame was not in a good state previously when repaired. Piet ordered a new A-frame. It will stand until new A-frame arrived. Lead

time is 60 days. We are expediting it to see if we can fly it in. My suggestion would be to remove A-frame and send to Epiroc for repairs. **(ongoing breakdown)**

A-FRAME: P/N 5590024183 **(77 days)**
SCREW HEX HEAD: P/N 5540423500 **(13 days)**
RETAINER: P/N 5590022045 **(85 days)**
HEX SOC SCREW: P/N 5750531559 **(13 days)**
SCREW HEX HEAD: P/N 0147110821 **(13 days)**
LOCK NUT: P/N 0291112832 **(13 days)**
WASHERS: P/N 5540113100 **(13 days)**

CAS L9 Implementation - Critical Issues Summary

- Invoicing and sign off to be completed.
- MH4.0 software to be loaded on the BEV machines. Asked for the signed documents to go ahead with installation. We need to do Lab scale tests with Strata. Waiting on feedback from them on when we can start with this.

Certiq & Mobilaris - Critical Issues Summary

Passport 360 Compliance

- BEV Technicians: **96%**
- CASL9 (Gloria): **99%**
- CASL9 (Nch2): **95%**
- CAS L9 (Nch3): **98%**
- Certiq & Mobilaris N2: **99%**
- Certiq & Mobilaris Gloria: **88%**

BEV Report

Weekly machine utilization summary:

Friday 09 January 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 – Afternoon shift: Electrical breakdown; Inverter units – No feedback. (3H05min)
4. FL 108 - Mechanical breakdown: Night shift – Changed a suction hose. **(12H42min)**
5. FL 112 -
6. FL 113 -
7. DT 146 - Night shift: Electrical breakdown – No feedback. **(8H56min)**

- Night shift: Auto electrical breakdown – Was standing for service that was not completed because auto was not available. (8H56min)
8. DT 147 -
 9. DT 149 -
 10. DT 150 - Night shift: Electrical breakdown; Inverter error – Was standing with auto phasing we did ID RUN on the machine, and it worked few hours, it was then reported again end off shift with same fault. (8H47min).
Morning shift: Electrical breakdown - Need to do fault finding. Cable damaged. (4H16min)
Night shift: Electrical breakdown – Busy fault finding. (4H16min)
 11. DT 162 - Morning shift: Auto electrical breakdown – Was standing with aircon issues (Aircon and bucket movement faults). (4H27min)
Afternoon shift: Electrical breakdown – Busy fault finding. (2H05min)
 12. DT 163 -
 13. DT 171 – Ongoing breakdown from 28/12/2025

Saturday 10 January 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 - Morning shift: Electrical breakdown – **No feedback.** (1H07min)
8. DT 147 - Morning shift: Electrical breakdown; Inverter – Replaced rear traction motor encoder cable and W2 HV cable. (1H07min)
9. DT 149 -
10. DT 150 -
11. DT 162 - Morning shift: 3rd party supplier fault – STRATA (1H07min)
Morning shift: Electrical breakdown; module error – D510 and D512 24V supply cable connector was loose From D522 I/O module's T-junction. (1H35min)
12. DT 163 -
13. DT 171 - Ongoing breakdown from 28/12/2025

Sunday 11 January 2026:

1. FL 98 -
2. FL 99 - Morning shift: Mechanical breakdown – Replaced charge pump. (1D03H01min)
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 -
10. DT 150 - Morning shift: Electrical breakdown – Standing with Auto phasing U801 fault. Front traction motor encoder cable damaged due to a hydraulic hose that was lying on top of the cable on the motor. Replaced cable. (3D04H08min)
11. DT 162 -

- 12. DT 163 -
- 13. DT 171 - Ongoing breakdown from 28/12/2025

Monday 12 January 2026:

- 1. FL 98 -
- 2. FL 99 -
- 3. FL 107 -
- 4. FL 108 - Morning shift: Mechanical breakdown – Changed transmission mounting rubbers. (04H35min)
- 5. FL 112 -
- 6. FL 113 -
- 7. DT 146 -
- 8. DT 147 -
- 9. DT 149 - Morning shift: 3rd party supplier fault – STRATA (1H35min)
- 10. DT 150 -
- 11. DT 162 - Morning shift: 3rd party supplier fault – STRATA (04H08min)
Afternoon shift: 3rd party supplier fault – STRATA (01D09H51min)
- 12. DT 163 -
- 13. DT 171 - Ongoing breakdown from 28/12/2025

Tuesday 13 January 2026:

- 1. FL 98 -
- 2. FL 99 -
- 3. FL 107 -
- 4. FL 108 -
- 5. FL 112 -
- 6. FL 113 -
- 7. DT 146 -
- 8. DT 147 -
- 9. DT 149 -
- 10. DT 150 -
- 11. DT 162 -
- 12. DT 163 -
- 13. DT 171 - Ongoing breakdown from 28/12/2025

Wednesday 14 January 2026:

- 1. FL 98 -
- 2. FL 99 - Morning shift: Mechanical breakdown - Gears
- 3. FL 107 -
- 4. FL 108 - Night shift: Tyre Bay breakdown – At tyre bay to replace tyres.
- 5. FL 112 - Night shift: Boilermaker breakdown – Half arrows. (9H12min)
- 6. FL 113 -
- 7. DT 146 - Night shift: Mechanical breakdown – Reset fire suppression and recovered machine to workshop. (10H15min)
Night shift: 3rd party supplier fault – Petroman (36min)

8. DT 147 -
9. DT 149 -
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 171 - Ongoing breakdown from 28/12/2025

Thursday 15 January 2026:

1. **FL 98** - Morning shift: Mechanical breakdown – VPY00011 on machine, fault cleared itself. (3H31min)
Night shift: Mechanical breakdown – Oil leak, no mechanic available.
2. **FL 99** -
3. **FL 107** - Morning shift: Mechanical breakdown – VPY00049 on machine, Loose HVIL plug from TMS to battery. (1H03min)
4. **FL 108** -
5. **FL 112** -
6. **FL 113** -
7. **DT 146** - Morning shift: Mechanical breakdown – Reset fire suppression. (1H22min)
8. **DT 147** -
9. **DT 149** - Morning shift: Tyre Bay breakdown – Tyre replacement. (5H28min)
10. **DT 150** -
11. **DT 162** -
12. **DT 163** - Afternoon shift: Fitter breakdown – Top up hydraulic oil level. (40min)
Afternoon shift: 3rd party supplier fault – STRATA (34min)
13. **DT 171** - Ongoing breakdown from 28/12/2025

Planned work for the coming week:

Improvement and actions:

1. We need to look at the data of the charger interrupts.
2. Post 3,6 and 7 CCS connector cables need to be replaced. (Cause of High pin temp)
Parts arrived, need to be installed
3. **Auxiliary motor spline grease** TSNB not done. We need to schedule the machines to complete this campaign. Grease already on site. This have been raised multiple times; I need BR to schedule the machines.
4. **Connector audits on Machines done.**
5. **Regeneration knob set to 100% completed.**

Services

Nchwaning 3

Maintenance Schedule Jan 2026				
Monday	Tuesday	Wednesday	Thursday	Friday
19-Jan-26	20-Jan-26	21-Jan-26	22-Jan-26	23-Jan-26
	DT0147		DT0148	
FL0103		FL0113		FL0108
HD0049	HD0050	HD0051	HD0052	HD0055

Battery and Charger Status Report

General summary

ST14 – B4 battery packs

- We currently have 10 x ST14 – B4 battery packs and 6 x ST14 machines operational underground. This is equal to the committed ratio of 1.6 batteries per machine.

MT42 – B5 battery packs

- We currently have 12 x MT42 – B5 battery packs operational underground. This is above the committed ratio of 1.6 batteries per machine.

160 kW Chargers

- Post 3,6 and 7 CCS connector cables need to be replaced. New 300A cables was order and received. Parts arrived and must still to be installed.

General

- There are more noted charger stops; this is being investigated with initial indications being worn out charger and CCS battery inlet connectors. DCDC campaign completed.

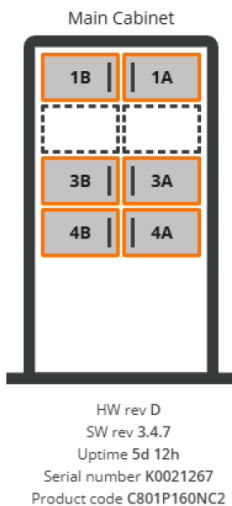
Table 1: Battery pack availability for week 2025

Battery Type	Battery ID	Status	Comment
B4 - ST14	VPY-00011	Working	
B4 - ST14	VPY-00031	Working	
B4 - ST14	VPY-00051	Breakdown	Not charging. The PSB board is defective. (5D)
B4 - ST14	VPY-00048	Working	
B4 - ST14	VPY-00049	Working	
B4 - ST14	VPY-00088	Working	

B4 - ST14	VPY-00086	Working	
B4 - ST14	VPY-00076	Working	
B4 - ST14	VPY-00083	Working	
B4 - ST14	VPY-00041	Working	
B5 - MT42	VPX-00016	Working	
B5 - MT42	VPX-00015	Working	
B5 - MT42	VPX-00023	Working	
B5 - MT42	VPX-00017	Working	
B5 - MT42	VPX-00010	Working	
B5 - MT42	VPX-00050	Working	
B5 - MT42	VPX-00031	Working	
B5 - MT42	VPX-00036	Working	
B5 - MT42	VPX-00028	Working	
B5 - MT42	VPX-00026	Working	09/01/26 - Low coolant pressure. Recharged coolant. (12H)
B5 - MT42	VPX-00024	Working	
B5 - MT42	VPX-00048	Working	
B5 - MT42	VPX-00044	Surface	On machine: 8997901500

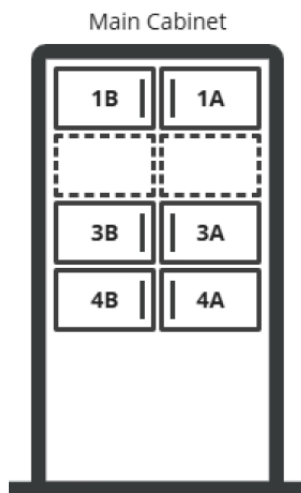
Battery charger summary:

Charger (BRM 1)



- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- When Charger modules are highlighted orange, Means charger is currently charging
- Charger currently on new SW V3.4.7

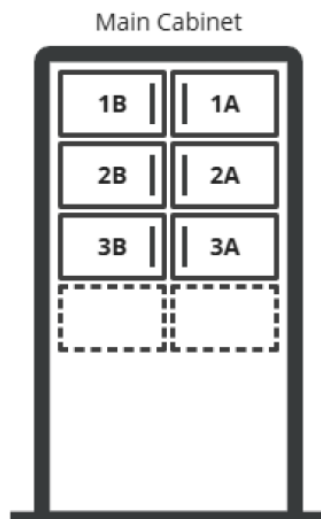
Charger (BRM 2)



HW rev D
SW rev 3.4.7
Uptime 17d 0h
Serial number K0021363
Product code C801P160NC2

- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- Charger currently on new SW V3.4.7

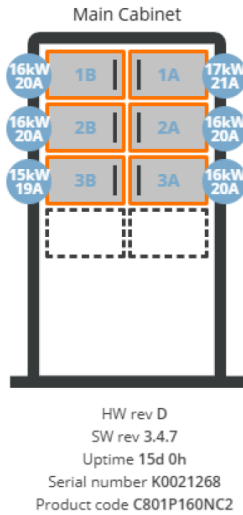
Charger (BRM 3)



HW rev D
SW rev 3.4.7
Uptime 9d 16h
Serial number K0021364
Product code C801P160NC2

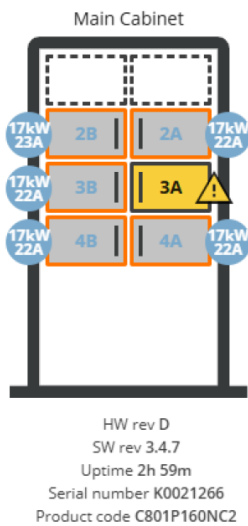
- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- Charger currently on new SW V3.4.7

Charger (BRM 4)



- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- When Charger modules are highlighted orange, Means charger is currently charging
- Charger currently on new SW V3.4.7

Charger (BRM 5)



ERRORS

POWER MODULE FAULT
MC 3A

- FAN_CURRENT
MC 3A
- RT_CONNECTOR_PMC_ERROR
MC 3A
- SSC_PMC_FAN_CURRENT
MC 3A

POWER MODULE FAULT
17.11.2025 13:32 (UTC+2)

PMC 3A on Main cabinet: SSC_PMC_FAN_CURRENT
Remote investigation required

POTENTIAL IMPACT
No information

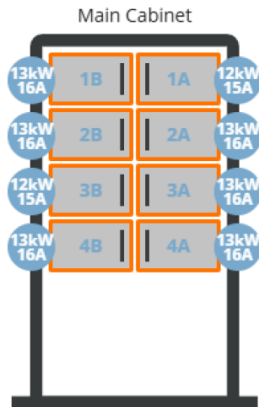
STEPS TO FIX
No information

DETAILS

Error code	CE_PMC_ERROR	Copy
Vendor	Kempower	
Occurred	17.11.2025 13:32 (UTC+2)	
Cause	No information	
Documentation	No information	

- Charger missing 2X power modules
- Charging power reduced
- Charger currently with a defective fan on the power module highlighted yellow, Charger fan needs to be replaced.
- Charger on new SW V3.4.7

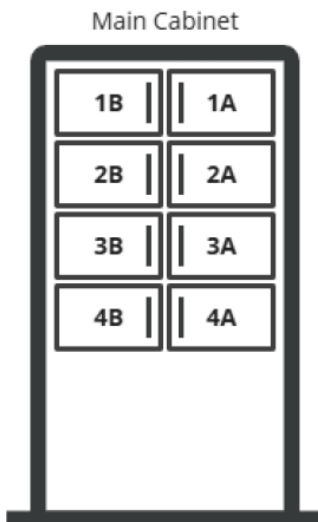
Charger (BRM 6)



- Charger currently running with no faults
- When Charger modules are highlighted orange, Means charger is currently charging
- Charger currently on new SW V3.4.7

HW rev D
SW rev 3.4.7
Uptime 9d 18h
Serial number K0021365
Product code C801P160NC2

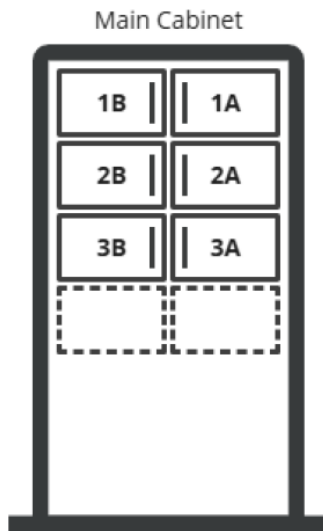
Charger (BRM 7)



- Charger currently running with no faults
- Charger currently on new SW V3.4.7

HW rev K
SW rev 3.4.7
Uptime 45m
Serial number K0028147
Product code C801P160ND4LC2

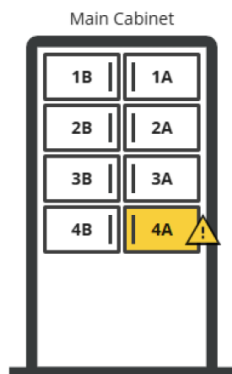
Charger (BRM 8)



HW rev K
SW rev 3.4.7
Uptime 12d 7h
Serial number K0028146
Product code C801P160ND4LC2

- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- Charger currently on new SW V3.4.7

Charger (BRM 9)



HW rev L
SW rev 3.4.7
Uptime 15d 11h
Serial number K0032320
Product code C801MVP160ND4LC2

ERRORS

▲ POWER MODULE FAULT
MC 4A

▲ RT_PMC_CONTINUOUS_FAILURES
MC 4A

RT_PMC_CONTINUOUS_FAILURES

28.11.2025 12:09 (UTC+2)

Power Module Channel (PMC) has continuous errors.
▲ Remote investigation required

POTENTIAL IMPACT

Charging power unit's total power output reduced by 25 kW.

STEPS TO FIX

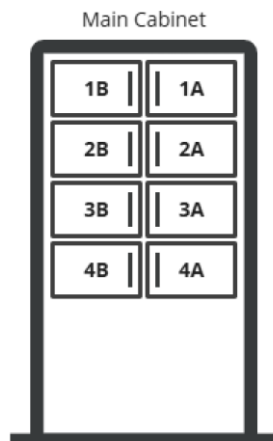
Do remote fault analysis with Service Partner or Kempower Technical Support.

DETAILS

Error code	RT_PMC_CONTINUOUS_FAILURES	Copy
Vendor	Kempower	
Occurred	28.11.2025 12:09 (UTC+2)	
Cause	This error appears after a runtime error has affected the Power Module channel (PMC) for more than 5 times by default.	
Documentation	No information	

- Charger with power module failure or module switched off
- Charger output reduced by 25KW
- Power module needs to be replaced if defective
- Charger on new SW version V3.4.7

Charger (BRM 10)



- Charger not connected to the Grid and offline
- Charger on old software and needs to be upgraded to SW V3.4.7

HW rev L
SW rev 2.4.20
Uptime 9m
Serial number K0032321
Product code
C801MVP160ND4LC2

FST Report Summary

Field Service on site:

CAS Report Summary

Report Type: Weekly
Start Date:
End Date:

Project Leader: Hennie van Niekerk
Report By:
Contract No: BPE145D

Critical Issues

Progress Update

All machines on Cas level 9.

Project in sign off phase.